

AIM: Develop Expense Tracking Module with Exception Handling

Source: File Name with extension
Traking.cs

Code:

```
using System;

class Program
{
    static void Main()
    {
        string[] category = new string[10];
        double[] amount = new double[10];
        string[] date = new string[10];

        int count = 0;
        int choice;

        do
        {
            Console.WriteLine("\n==== Expense Tracker =====");
            Console.WriteLine("1. Add Expense");
            Console.WriteLine("2. View Expenses");
            Console.WriteLine("3. Total Expense");
            Console.WriteLine("4. Exit");
            Console.Write("Enter your choice: ");

            try
            {
                choice = Convert.ToInt32(Console.ReadLine());

                switch (choice)
                {
                    case 1:
                        try
                        {
                            Console.Write("Enter Category: ");
                            category[count] = Console.ReadLine();

                            if (category[count] == "")
                                throw new Exception("Category cannot be empty.");

                            Console.Write("Enter Amount: ");
                            amount[count] = Convert.ToDouble(Console.ReadLine());

                            if (amount[count] <= 0)
                                throw new Exception("Amount must be greater than zero.");
                        }
                        catch { }
                    case 2:
                    case 3:
                    case 4:
                        break;
                }
            }
            catch { }

            count++;
        } while (choice != 4);
    }
}
```

```
        Console.WriteLine("Enter Date: ");
        date[count] = Console.ReadLine();

        count++;

        Console.WriteLine("Expense Added Successfully.");
    }
    catch (Exception ex)
    {
        Console.WriteLine("Error: " + ex.Message);
    }
    break;

case 2:
    if (count == 0)
    {
        Console.WriteLine("No expenses found.");
    }
    else
    {
        Console.WriteLine("\nExpense Details");
        for (int i = 0; i < count; i++)
        {
            Console.WriteLine("-----");
            Console.WriteLine("Category : " + category[i]);
            Console.WriteLine("Amount   : " + amount[i]);
            Console.WriteLine("Date     : " + date[i]);
        }
    }
    break;

case 3:
    double total = 0;

    for (int i = 0; i < count; i++)
    {
        total += amount[i];
    }

    Console.WriteLine("Total Expense = " + total);
    break;

case 4:
    Console.WriteLine("Thank You!");
    break;

default:
    Console.WriteLine("Invalid Choice");
    break;
}
}
catch (FormatException)
{
    Console.WriteLine("Please enter a valid number.");
    choice = 0;
}
```

```
    }  
    } while (choice != 4);  
}  
}
```

Output:

```
===== Expense Tracker =====  
1. Add Expense  
2. View Expenses  
3. Total Expense  
4. Exit  
Enter your choice: 1  
Enter Category: 2  
Enter Amount: 10000  
Enter Date: 12/05/2027  
Expense Added Successfully.  
  
===== Expense Tracker =====  
1. Add Expense  
2. View Expenses  
3. Total Expense  
4. Exit  
Enter your choice: 2  
  
Expense Details  
-----  
Category : 2  
Amount   : 10000  
Date      : 12/0/2027  
  
===== Expense Tracker =====  
1. Add Expense  
2. View Expenses  
3. Total Expense  
4. Exit  
Enter your choice: 3  
Total Expense = 10000
```